Area and Perimeter: Mixed Practice

Name _____ Date _____ Calculator allowed Version 1 Scan to practise this topic online

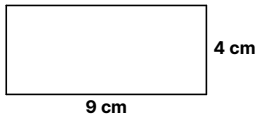
KEY WORDS area *(horahanga)* · perimeter *(paenga)* · circumference *(paenga porowhita)* · hectare *(heketea)*

A Choose the formula ●○○

Remember: Name the shape first, then write its area formula before you put the numbers in.

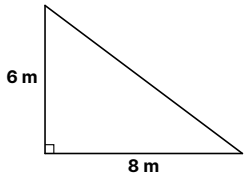
1. Find the area of each shape. Give circle answers to 1 decimal place.

a)



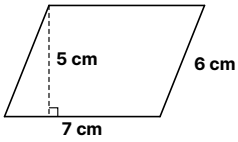
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b)



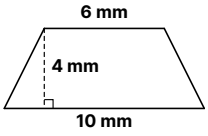
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c)



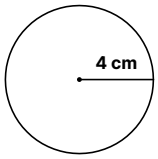
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d)



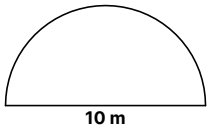
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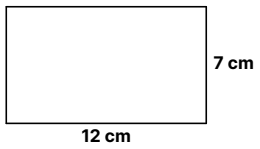
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B Perimeter ●●○

Remember: Perimeter is the distance around the outside: add every edge. For a curved edge use the circumference formula, and do not forget the straight edges of a semicircle or quarter circle.

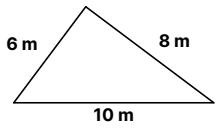
2. Find the perimeter of each shape. Give answers with π to 1 decimal place.

a)



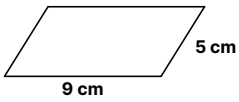
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b)



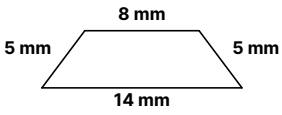
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c)



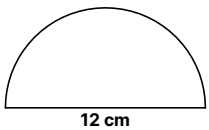
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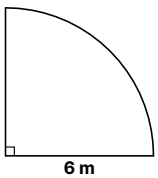
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C Working backwards ●●○

Remember: Write the formula, put in what you know, and undo the operations one at a time.

3. Find the missing length.

a) A square has an area of 144 cm². Find its side length.

b) A rectangle has an area of 84 m² and a length of 12 m. Find its width.

c) A triangle has an area of 36 cm² and a base of 9 cm. Find its height.

d) A circle has a circumference of 50 cm. Find its radius, to 1 decimal place.

e) A circle has an area of 200 m². Find its radius, to 1 decimal place.

f) A trapezium has parallel sides of 9 cm and 6 cm and an area of 60 cm². Find its height.

D Units of area ●●○

Remember: 1 cm² = 100 mm², 1 m² = 10 000 cm², 1 hectare (ha) = 10 000 m², 1 km² = 100 ha. A length unit is multiplied by 10 or 100, so the area unit is multiplied by 10² or 100².

4. Change the units.

a) Change 3.5 cm² to mm².

b) Change 2.4 m² to cm².

c) Change 7 ha to m².

d) Change 850 mm² to cm².

e) Change 45,000 cm² to m².

f) Change 3.2 km² to ha.

E Problems ●●●

5. A rectangular section is 18 m by 12 m.

a) Fencing costs $24 per metre. Find the cost of fencing all four sides.

b) Lawn seed covers 1 m² for $7.5. Find the cost of seeding the whole section.

6. A round trampoline has a diameter of 3.6 m.

a) Find the area of the jumping mat, to 2 decimal places.

b) Safety padding goes right around the edge and costs $85 per metre. Find the cost of the padding, to the nearest dollar.

7. A triangular sail has a base of 4.5 m, a height of 6 m and a longest side of 7.5 m. Rope is sewn along all three edges. The third edge is as long as the height. How much rope is needed?

8. A paddock is a rectangle 400 m by 350 m. Find its area in hectares.

★ Extension

9. A rectangle has a perimeter of 34 cm and an area of 60 cm². Find its length and width.

10. A square and a circle each have a perimeter of 40 cm. How much bigger is the area of the circle than the area of the square? Give the answer to 1 decimal place.

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Area and Perimeter: Mixed Practice

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

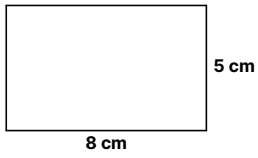
KEY WORDS area *(horahanga)* · perimeter *(paenga)* · circumference *(paenga porowhita)* · hectare *(heketea)*

A Choose the formula ●○○

Remember: Name the shape first, then write its area formula before you put the numbers in.

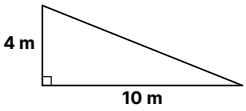
1. Find the area of each shape. Give circle answers to 1 decimal place.

a)



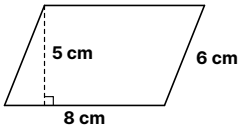
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b)



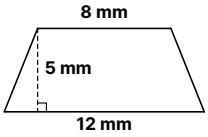
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c)



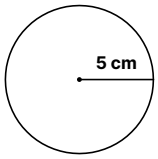
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d)



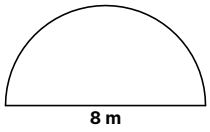
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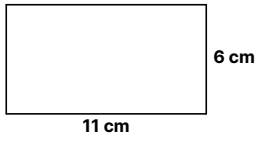
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B Perimeter ●●○

Remember: Perimeter is the distance around the outside: add every edge. For a curved edge use the circumference formula, and do not forget the straight edges of a semicircle or quarter circle.

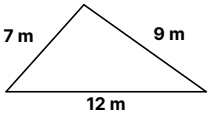
2. Find the perimeter of each shape. Give answers with π to 1 decimal place.

a)



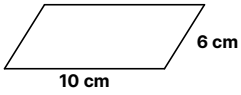
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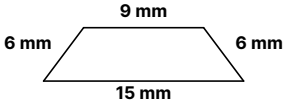
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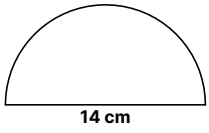
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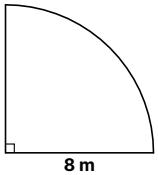
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f)



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C Working backwards ●●○

Remember: Write the formula, put in what you know, and undo the operations one at a time.

3. Find the missing length.

a) A square has an area of 169 cm². Find its side length.

b) A rectangle has an area of 96 m² and a length of 12 m. Find its width.

c) A triangle has an area of 40 cm² and a base of 8 cm. Find its height.

d) A circle has a circumference of 60 cm. Find its radius, to 1 decimal place.

e) A circle has an area of 250 m². Find its radius, to 1 decimal place.

f) A trapezium has parallel sides of 10 cm and 4 cm and an area of 70 cm². Find its height.

D Units of area ●●○

Remember: 1 cm² = 100 mm², 1 m² = 10 000 cm², 1 hectare (ha) = 10 000 m², 1 km² = 100 ha. A length unit is multiplied by 10 or 100, so the area unit is multiplied by 10² or 100².

4. Change the units.

a) Change 4.5 cm² to mm².

b) Change 1.8 m² to cm².

c) Change 6 ha to m².

d) Change 640 mm² to cm².

e) Change 72,000 cm² to m².

f) Change 2.6 km² to ha.

E Problems ●●●

5. A rectangular section is 20 m by 14 m.

a) Fencing costs $26 per metre. Find the cost of fencing all four sides.

b) Lawn seed covers 1 m² for $8. Find the cost of seeding the whole section.

6. A round trampoline has a diameter of 4.2 m.

a) Find the area of the jumping mat, to 2 decimal places.

b) Safety padding goes right around the edge and costs $90 per metre. Find the cost of the padding, to the nearest dollar.

7. A triangular sail has a base of 5.2 m, a height of 6 m and a longest side of 7.9 m. Rope is sewn along all three edges. The third edge is as long as the height. How much rope is needed?

8. A paddock is a rectangle 450 m by 300 m. Find its area in hectares.

★ Extension

9. A rectangle has a perimeter of 22 cm and an area of 24 cm². Find its length and width.

10. A square and a circle each have a perimeter of 36 cm. How much bigger is the area of the circle than the area of the square? Give the answer to 1 decimal place.

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